ENGINEERING & DATA SCIENCES · CHEMISTRY S1-1

Exam practice

The atom and its quantum model

Handout chapters 1–3

25 questions · Moderate and Hard ★ only

How to use it

- The questions cover every type of exercise that chapters 1–3 allow: calculations, identifications, justifications, data analysis and multi-step problems. Easy recall questions are left out; the worksheets cover them.
- Work on paper with a calculator, then check with *Chem-S1-1_Exam-Practice-Ch1-3_Answers.pdf*.
- Moderate: one or two steps with a formula from the handout. Hard: several steps, reasoning in reverse, or ideas from two chapters at once.

Data $c = 3.00 \times 10^{8}\,\mathrm{m/s}$ $h = 6.62 \times 10^{-34}\,\mathrm{J\cdot s}$ $e = 1.6 \times 10^{-19}\,\mathrm{C}$ $1\ \mathrm{eV} = 1.6 \times 10^{-19}\,\mathrm{J}$ $N_A = 6.022 \times 10^{23}/\mathrm{mol}$ $R_H = 3.29 \times 10^{15}\,\mathrm{Hz}$ $a_0 = 52.9\ \mathrm{pm}$ $k = 1/(4\pi\varepsilon_0) = 8.99 \times 10^{9}\,\mathrm{N\cdot m^2/C^2}$ $m_e = 9.1094 \times 10^{-31}\,\mathrm{kg}$ $m_p = 1.0073\ \mathrm{amu}$ $m_n = 1.0087\ \mathrm{amu}$ $1\ \mathrm{amu} = 1.6605 \times 10^{-27}\,\mathrm{kg}$ $g = 9.81\,\mathrm{m/s^2}$ $E_n = -13.6\,Z^2/n^2\ \mathrm{eV}$.

Map of the question types

Chapter	Type of question	Questions
1	Counting protons, neutrons, electrons; isotopes and isoelectronic species	A1
1	Mass defect and energy of a nucleus	A2
1	Millikan: finding e from data; balancing a charged drop	A3
1	Rutherford: closest approach, density and size of the nucleus	A4
1	Thomson: comparing charge-to-mass ratios	A5
2	Converting between λ, ν, wavenumber, eV and $\mathrm{kJ\,mol^{-1}}$	B1, B7
2	Identifying a transition from a wavelength	B2
2	Which photons are absorbed; ionisation and the ejected electron	B3, D3
2	Emission cascade: number of lines, extreme wavelengths, visible lines	B4
2	Data analysis: Rydberg's constant from measured lines	B5
2	Hydrogen-like ions; coinciding lines	B6, D2
2	Explaining spectra (lines vs bands, emission vs absorption, colour)	B8
3	Allowed sets of quantum numbers; naming orbitals	C1
3	Counting orbitals and electrons with conditions	C2
3	Nodes; recognising an orbital from its nodes and planes	C3, C8
3	Radial probability density: maximum, probability inside a sphere	C4, C5
3	Degeneracy in H vs many-electron atoms; Madelung order	C6
3	Spin (sodium doublet) and Zeeman splitting	C7
3	Sizes $r = n^2 a_0/Z$	C9, D2
1–3	Synthesis: Coulomb energy and Bohr levels; unknown ion; photoionisation	D1–D3

Part A · The atom and its particles (chapter 1)

A1. Moderate Particle bookkeeping

(a) Complete the table.

Species	$^{37}_{17}\mathrm{Cl}^-$	$^{56}_{26}\mathrm{Fe}^{3+}$	$^{18}_{8}\mathrm{O}^{2-}$	$^{16}_{8}\mathrm{O}$	$^{23}_{11}\mathrm{Na}^+$
protons / neutrons / electrons					

(b) Which species are isotopes of each other? Which are isoelectronic, and with which noble gas? (c) A species has 26 protons, 30 neutrons and 23 electrons. Write its symbol in full.

A2. Hard ★ Where does the mass go? (helium-4)

A ^{4}He atom has a mass of 4.0026 amu. (a) Compute the total mass of its separate particles (2 protons, 2 neutrons, 2 electrons; $m_e = 1/1823$ amu). (b) Deduce the mass defect Δm in amu and in kg. (c) The binding energy is $E = \Delta m\, c^2$: compute it per atom, in J and MeV (1 MeV $= 1.6 \times 10^{-13}$ J), and per mole. Compare with a chemical bond (a few hundred kJ mol^{-1}).

A3. Hard ★ Millikan's experiment

(a) Charges measured on oil drops: 6.41, 9.62, 3.20, 12.82 and 4.81×10^{-19} C. Show that they are whole multiples of one charge and find it.
(b) A drop of oil (density 886 kg m^{-3}, radius 1.00 µm) is held still between two horizontal plates $d = 5.0$ mm apart when $U = 380$ V is applied: the electric force qU/d balances the weight mg. Find its charge and the number of extra electrons it carries.

A4. Hard ★ Rutherford on copper

An α particle (charge $+2e$) with kinetic energy 7.7 MeV heads straight for a copper nucleus ($Z = 29$). (a) At its closest approach all its kinetic energy has become potential energy $k\,q_1 q_2/d$. Find d. (b) Why is d only an upper limit for the radius of the nucleus? (c) The copper nucleus ($A = 63$) has a radius of about 4.8 fm. Compute its density and compare the radius of the atom (128 pm) with that of its nucleus. (d) Use this ratio to explain why most α particles cross a metal foil undeflected.

A5. Moderate Thomson's specific charge

In a deflection tube, the deviation is proportional to q/m. Compute q/m for an electron, a proton and an α particle ($m_\alpha = 4.0015$ amu) and rank their deviations. By what factor is the electron deviated more than the α particle?

Part B · Light and the hydrogen spectrum (chapter 2)

B1. Moderate One photon, five units

A photon carries 2.48 eV. Give its wavelength, frequency, wavenumber (cm^{-1}), the energy of one mole of such photons (kJ mol^{-1}), and the region of the spectrum.

B2. Moderate Which transition?

Hydrogen emits a line at $\lambda = 1282$ nm. Using Rydberg's formula, show that it cannot belong to the Lyman or the Balmer series, and identify the transition.

B3. Moderate Photons hitting ground-state hydrogen

Hydrogen atoms in their ground state receive photons of (a) 12.75 eV, (b) 12.50 eV, (c) 13.22 eV, (d) 20.0 eV. For each, say whether it is absorbed and what happens. For (d), compute the speed of the electron that leaves.

Unofficial study material

B4. Hard ★ Emission cascade from $n = 5$

A gas of hydrogen atoms is excited to $n = 5$. (a) How many different lines can it emit? (b) How many of them are visible, and which? (c) Compute the shortest and the longest wavelengths emitted. (d) Draw the level diagram with the transitions, grouped into series.

B5. Hard ★ Measuring Rydberg's constant

The four visible lines of hydrogen are measured at 656.3, 486.1, 434.0 and 410.2 nm. (a) Show that Balmer's formula predicts a straight line when $1/\lambda$ is plotted against $1/n^2$, and give its slope and intercept. (b) Using a linear fit (or the first and last points), find R_H in m^{-1} and in Hz. (c) Deduce the ionisation energy of hydrogen in eV and $\mathrm{kJ\,mol^{-1}}$, and the wavelength of the Balmer limit.

B6. Hard ★ The Li^{2+} ion

(a) Compute the ionisation energy of Li^{2+} in its ground state and the wavelength of its $2 \to 1$ line. In which region is it? (b) Show that the Li^{2+} transition $9 \to 6$ has exactly the wavelength of hydrogen's $3 \to 2$ line. (c) Generalise: which Li^{2+} transitions reproduce all the Balmer lines?

B7. Moderate Counting photons

A 60 W sodium lamp turns 30 % of its power into light at 589 nm. How many photons does it emit per second, and how many moles of photons per hour?

B8. Moderate Explaining spectra

(a) Why do the absorption and emission spectra of hydrogen show lines at exactly the same wavelengths? (b) Why does a solution of $KMnO_4$ show a broad absorption band around 530 nm rather than thin lines, and why does it look purple? (c) What experimental fact led Bohr to say that the energy of the atom is quantised?

Part C · Quantum numbers and atomic orbitals (chapter 3)

C1. Moderate Allowed or not?

For each set (n, ℓ, m_ℓ, m_s), say whether it is allowed. If so, name the orbital; if not, say which rule is broken: (a) $(2, 2, 0, +\tfrac{1}{2})$; (b) $(3, 2, -2, -\tfrac{1}{2})$; (c) $(4, 0, 1, +\tfrac{1}{2})$; (d) $(5, 3, -3, +\tfrac{1}{2})$; (e) $(1, 0, 0, 0)$; (f) $(3, 1, -1, +\tfrac{1}{2})$.

C2. Moderate Counting with conditions

(a) For $n = 4$: list the subshells and give the numbers of orbitals and electrons. (b) How many electrons of an atom can have $n = 3$ and $m_\ell = +1$? (c) How many can have $n = 4$ and $m_s = +\tfrac{1}{2}$? (d) How many orbitals with $\ell = 2$ exist for $n \le 5$?

C3. Moderate Nodes

(a) Give the numbers of radial, angular and total nodes of 2s, 3p, 4d, 5f and 4s. (b) Which orbital has 2 radial nodes and 1 angular node? Which has no radial node and 3 angular nodes? (c) Can an orbital of the shell $n = 4$ have 2 radial and 2 angular nodes?

C4. Hard ★ Where is the 1s electron?

For hydrogen, $P(r) = \dfrac{4}{a_0^3}\, r^2\, e^{-2r/a_0}$. (a) Show that P is maximum at $r = a_0$. (b) Compute $P(2a_0)/P(a_0)$. (c) Show that the probability of finding the electron inside the sphere of radius a_0 is $1 - 5e^{-2} \approx 32\,\%$ (use $\int x^2 e^{-x}\,\mathrm{d}x = -(x^2 + 2x + 2)\,e^{-x}$). (d) Where is the maximum for He^+?

C5. Hard ★ The 2s orbital and its node

For the hydrogen 2s orbital, $\Psi_{2s} \propto \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$. (a) Find the radius of the radial node, in pm. (b) The radial density is $P(r) \propto r^2 \left(2 - \frac{r}{a_0}\right)^2 e^{-r/a_0}$: show that its maxima are at $r = (3 \pm \sqrt{5})\, a_0$, and say which one is the most probable distance. (c) Where is the node for Li^{2+}?

C6. Moderate Degeneracy and Madelung's order

(a) In hydrogen, how many orbitals have the energy E_3? Give their energy in eV. (b) Why does this degeneracy disappear in a many-electron atom such as Na? Order 3s, 3p, 3d for Na. (c) Use Madelung's rule to order 6s, 4f, 5d, 6p, 5f, 7s, and say which pairs have the same $n + \ell$.

C7. Moderate Spin and magnetic field

(a) Sodium's D line is a doublet at 589.0 and 589.6 nm. Compute the energy gap between the two upper levels in meV and the wavenumber gap in cm^{-1}. What property of the electron explains it? (b) Into how many sublevels does a 3d level split in a magnetic field (Zeeman effect)? A 4f level? Which quantum number is revealed?

C8. Moderate Recognise the orbital

Name the orbital: (a) one nodal plane, the (xOy) plane, and no radial node; (b) two angular nodes and one radial node; (c) spherical, with two radial nodes; (d) $\Psi_{3,1,0}$. For (a) and (d), give m_ℓ and the occupied/unoccupied status in ground-state hydrogen.

C9. Hard ★ Sizes of hydrogen-like orbitals

Using $r = n^2 a_0 / Z$: (a) compare the sizes of H 1s, Li^{2+} 2s and Li^{2+} 3d. (b) Which hydrogen-like ion has a 2p orbital the same size as the 1s of hydrogen? Give the energy of that 2p level.

Part D · Synthesis: chapters 1 to 3 together

D1. Hard ★ Coulomb meets Bohr

In the Bohr picture the electron of hydrogen orbits at $r_n = n^2 a_0$. Its total energy is $E = -\frac{ke^2}{2r}$ (half its Coulomb potential energy). (a) Compute E for $r = a_0$ in J and eV and compare with E_1. (b) Show that this expression gives $E_n = E_1 / n^2$. (c) For He^+ in $n = 2$, give r and E; compare with hydrogen's ground state.

D2. Hard ★ An unknown ion

A hydrogen-like ion emits a series ending on $n_1 = 2$ whose limit is at 22.8 nm. Identify the ion, then give its ground-state ionisation energy and the size of its 1s orbital.

D3. Hard ★ Photoionisation by a helium lamp

A helium discharge lamp emits UV light at 58.4 nm. (a) Compute the photon energy in eV. (b) It hits ground-state hydrogen atoms: what happens, and what is the kinetic energy and speed of the electron that leaves? (c) Could this lamp ionise He^+? Justify.